

Question 1. (calculation)

- a) Let $a(t) = \delta(t)$. What is its Fourier Series set of coefficients, $A(\omega)$? Make plots of both the phase and magnitude spectra.
- b) Let $b(t) = \delta(t - 3)$. What is its Fourier Series set of coefficients, $B(\omega)$? Make plots of both the phase and magnitude spectra.
- c) Substitute the values of $A(\omega)$ and $B(\omega)$ into each of their respective synthesis equations. What do you notice about each of their integrals? Can they be simplified at all (hint: check carefully to make sure any steps to simplify are valid)?

Question 2. (calculation)

Let $f(t) = e^{-a|t|}$. What are the set of Fourier coefficients, $F(\omega)$? Hint: you will want to notice some even/odd considerations to start. You will probably have to use integration-by-parts, as well; as this proceeds it may look like you get stuck in a circular loop, but keep a look out for integrals that look similar pieces for what you have.

Question 3. (programming)

In class, we will have gone through the Fourier Transform of this function: $g(t) = (1/\sqrt{2\pi\sigma^2}) e^{-0.5(t/\sigma)^2}$. Similar to what we did in the tutorial this week, plot $g(t)$ and $|G(\omega)|$ next to each other. In this case, make plots with $\sigma = 0.5, 1, 2, 4$. When you plot the time values, use a constant xlim and ylim across those plots; and when you are plotting the magnitude spectrum, define some constant xlim and ylim values for those (the limits need not be the same for the time and frequency plots).